Hybrid Solar/Wind Digital Twin: Model Explanation

Project Goal

Create a synthetic digital-twin dataset for a hybrid Solar/Wind system and train baseline and forecast AI models for power estimation.

Pipeline Outputs

1. energy_dataset.csv (8760 hourly rows)
2. Static plots in plots folder
3. Live dashboard plots during execution (optional)
4. Baseline model artifact: power_model_rf.joblib
5. Forecast model artifact: power_forecast_1h_rf.joblib

Data Synthesis

- Lux follows a diurnal sine pattern, seasonal modulation, and cloud noise.
- Wind speed is sampled from a Weibull distribution.
- Temperature combines annual cycle, daily cycle, lux correlation, and noise.
- Power combines solar and wind terms; current uses I = P / V.

Physics-Inspired Equations

```
P_solar = P_rated * (lux / lux_max)
P_wind = k * v^3
P_total = P_solar + P_wind
I = P / V
```

Model Training

- Baseline model: RandomForestRegressor predicts same-hour power_produced_w.
- Forecast model: RandomForestRegressor predicts next-hour power_next_1h_w.
- Baseline split is random 80/20; forecast split is chronological 80/20.
- Metrics: MAE (W) and R2.

Metric Interpretation

MAE is average absolute error in watts. For hourly data, MAE in watts is approximately equivalent to Wh error per hour interval.

How To Run

```
.\.venv\Scripts\python.exe .\train_energy_model.py
```

```
.\.venv\Scripts\python.exe .\train_energy_model.py --no-live-plots
```